Web Services, Google, IBM, Microsoft, Oracle, Salesforce, SAP, Teradata, Alibaba, and Tencent.

According to an Allied Market Research report, the global hybrid cloud market was valued at $96.7 billion in 2023, and is estimated to reach $414.1 billion by 2032, growing at a CAGR of 17.2% from 2024 to 2032.

Research conducted by Foundry shows that 98% of IT leaders have adopted or plan to adopt a hybrid model specifically to balance AI performance. The report highlights that generative AI applications in co-location centres rose from 42% to 45% year-over-year as organisations prioritised low latency and data control over cloud-native convenience.

A recent report from Gartner stated that sovereign cloud infrastructure spending will leap 35.6% year-over-year to $80 billion in 2026. They estimate that 20% of cloud infrastructure workloads will shift from global hyperscales to local/regional providers this year to meet digital independence goals.

What enterprises should consider before opting for the hybrid cloud

A hybrid cloud plays a key role in increasing the speed of delivery of IT resources to end users, improves disaster recovery capabilities, and enables better resource utilisation. However, organisations should consider a few things before going for a hybrid cloud.

Visibility of current state: The current application landscape and infrastructure across the enterprise must be assessed. A complete application portfolio analysis of the enterprise should be performed to decide on the cloud adoption.

Rate of cloud adoption: A big bang approach will not work. The timelines of cloud adoption should be based on the criticality of the applications. The factors that decide the rate of cloud adoption include complexity of the applications, data requirements, regulatory and compliance needs, modernisation prerequisites, cost implications, and real-time requirements.

Portfolio rationalisation: The business functions and the applications that are catering to them need to be identified and re-engineered based on industry trends and mergers/acquisitions. Redundant functionalities across applications must be rationalised before moving to the hybrid cloud.

Application migration: Applications can be migrated to and from the data centre and cloud using the hybrid cloud model. The applications that need to remain on-premises or move to a private or public cloud must be identified. This migration can be temporary or permanent depending on the strategy of migration.

Nature of applications: Applications that change frequently must be moved to the cloud to leverage automated deployment through DevOps. Applications that handle sensitive data are best retained on-premises. Applications with very high scalability requirements because of varied user load are ideal to be hosted on the public/private cloud.

Selection of environments: Public cloud environments may not provide specialised hardware. Choose the best environment for the application to run to deliver the functionality at the most optimal cost and effort.

Integration strategy: There is a need to connect the applications back to the historical data that resides on the on-premises servers even after the cloud migration. Enterprises must develop an integration strategy to be followed by the hybrid cloud.

Regulatory requirements: Applications requiring regulatory and compliance requirements demand some of the applications and data to reside on-premises. This requires due diligence to select the right candidates for the hybrid cloud.

Containerisation: Containerisation helps make the application cloud-agnostic and move across public, private, and on-premises clouds.

Cloud interoperability: Integration between several cloud offerings across multiple cloud service providers and cloud types is a key consideration for the success of hybrid cloud adoption.

Hybrid cloud management model

The hybrid cloud adoption management model is depicted in Figure 1. The key layers of this model are:

Unified channels and edge devices: This is the user interface and access layer through which users interact with the cloud system, such as the web, mobile, and IoT devices.

Agentic interfaces*:* These serve as the interface for autonomous AI agents that can now interact with users like a human collaborator.

Modernised enterprise application portfolio: This layer categorises the applications that run on the hybrid cloud into microservices (modern), monoliths (traditional), and legacy (old systems).

Cloud-native microservices: These are applications designed as small, independent services for easier updates and scaling.

Agentic services: These autonomous workflows (using frameworks like LangGraph) can execute complex multi-step tasks across different business apps.

Integrated service fabric: This layer enables different components of the platform to communicate with each other effectively and securely. It includes:

- *Open-source API gateway (e.g., Kong, Envoy):* Centralises API management, security, and traffic control. Used to route requests to the correct model or microservice.
- *Cloud-native service mesh (e.g. Istio, Linkerd):* Manages service-to-service communication, providing advanced features like intelligent routing, encryption (mTLS), and